Kindergarten Grade-Level Expanded Expectations

NUMBER SENSE: NS

K.NS.A	**Know number names and count sequence**	
K.NS.A.1	Count to 100 by ones and tens.	The expectation of the student is to count verbally to one hundred by ones and tens. The use of counting by tens strategy should support the counting by ones. Rote counting is an initial step to understanding the base-ten number system.
K.NS.A.2	Count forward beginning from a given number between 1 and 20.	The expectation of the student is to count forward within 20 verbally beginning from a given number (instead of having to begin at 1). There should be specific focus on the numbers in the range from eleven to twenty.
K.NS.A.3	Count backward from a given number between 10 and 1.	The expectation of the student is to count backward from a given (or student generated) number between ten and one. This may be used as an initial strategy for subtraction.
K.NS.A.4	Read and write numerals and represent a number of objects from 0 to 20.	The expectation of the student is to read and write numbers from zero to twenty, and represent a number of objects with a written numeral from zero to twenty (with zero representing a count of no objects).
K.NS.B	**Understand the relationship between numbers and quantities; connect counting to cardinality.**	
K.NS.B.5	Say the number names when counting objects, in the standard order, pairing each object with one and only one number name and each number name with one and only one object.	The expectation of the student is to demonstrate a one-to-one correspondence when counting objects. Say the number names in the standard order, pairing each object with one and only one number name and each number name with one and only one object, connect rote counting to cardinality. (limited to twenty objects) Demonstrate one-to-one correspondence when counting objects.

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K.NS.B.6	Demonstrate that the last number name said tells the number of objects counted and the number of objects is the same regardless of their arrangement or the order in which they were counted.	The expectation of the student is to demonstrate that the last number name said when counting tells the number of objects counted. The number of objects is the same regardless of their arrangement or the order in which they were counted. The use of manipulatives is vital. (limited to twenty objects)
K.NS.B.7	Demonstrate that each successive number name refers to a quantity that is one larger than the previous number.	The expectation of the student is to demonstrate that each successive number name refers to a quantity that is one larger than the previous number. (limited to twenty)
K.NS.B.8	Recognize, without counting, the quantity of groups up to 5 objects arranged in common patterns.	The expectation of the student is to subitize a quantity. Without counting, recognize the quantity of groups up to five objects arranged in common patterns (e.g. dice, dominoes, five frames, playing cards, ten frames, dot cards, etc.).
K.NS.B.9	Demonstrate that a number can be used to represent “how many” are in a set.	The expectation of the student is to demonstrate that a number can be used to represent “how many” are in a set (cardinality). (limited to twenty) Given a number from one to twenty, count out that many objects.
K.NS.C	**Compare numbers.**	
K.NS.C.10	Compare two or more sets of objects and identify which set is equal to, more than or less than the other.	The expectation of the student is to compare two or more sets of objects (up to ten objects in each group) and identify which set is equal to, more than or fewer (less) than the other.
K.NS.C.11	Compare two numerals, between 1 and 10, and determine which is more than or less than the other.	The expectation of the student is to compare two written numerals, each between 1 and 10, and determine which represents a quantity that is more than or less than the other.
NUMBER SENSE AND OPERATIONS IN BASE TEN: NBT		
K.NBT.A	**Work with numbers 11 – 19 to gain foundations for place value.**	

K.NBT.A.1	Compose and decompose numbers from 11 to 19 into sets of tens with additional ones.	The expectation of the student is to compose and decompose numbers from 11 to 19 into sets of tens with additional ones. Separate ten ones from the remaining ones, using manipulatives or drawings. The student should demonstrate (verbally and symbolically) an understanding that the numbers from eleven to nineteen are composed of a group of ten and additional one(s). Compose and decompose numbers from 11 to 19 into sets of tens with additional ones (separating ten ones from the remaining ones) by using objects or drawings. Understand that these numbers are composed of one group of ten and one, two, three, four, five, six, seven, eight or nine ones.
RELATIONSHIPS AND ALGEBRAIC THINKING: RA		
K.RA.A	**Understand addition as putting together or adding to, and understand subtraction as taking apart or taking from.**	
K.RA.A.1	Represent addition and subtraction within 10.	The expectation of the student is to use a variety of strategies to represent sums and differences within ten. (e.g., objects, fingers, mental images, drawings, sounds, acting out situations, verbal explanations, expressions or equations) *No single strategy is recommended over another. Consider the needs of the student.*
K.RA.A.2	Demonstrate fluency for addition and subtraction within 5.	The expectation of the student is to demonstrate fluency for addition and subtraction within 5. (*Fluency refers to accuracy and efficiency and does not equate to memorization.)*
K.RA.A.3	Decompose numbers less than or equal to 10 in more than one way.	The expectation of the student is to decompose (compose) numbers to ten using methods such as using objects or drawings, and record each result using a drawing or equation. (e.g., 2 + 3 = 5 and 5 = 4 +1)
K.RA.A.4	Make 10 for any number from 1 to 9.	The expectation of the student is to use various strategies (objects, mentally, etc.) to make 10 and to model the work with a drawing or equation. For any number from 1 to 9, find the number that makes 10 when added to the given number.
GEOMETRY AND MEASUREMENT: GM		
K.GM.A	**Reason with shapes and their attributes.**	

K.GM.A.1	Describe several measureable attributes of objects.	The expectation of the student is to describe several measureable attributes of an object, using appropriate language (e.g., length, weight, height or capacity).
K.GM.A.2	Compare the measurable attributes of two objects.	The expectation of the student is to compare the measureable attributes of two objects, using appropriate language. (e.g., longer, taller, shorter, same length, heavier, lighter, same weight, holds more, holds less, holds the same amount, etc.)
K.GM.B	**Work with time and money.**	
K.GM.B.3	Demonstrate an understanding of concepts of time and devices that measure time.	The expectation of the student is to demonstrate an understanding of the concepts of time (e.g., morning, afternoon, evening, today, yesterday, tomorrow, week and year) and tools that measure time. (e.g., clock or calendar)
K.GM.B.4	Name the days of the week.	The expectation of the student is to verbally name the days of the week by rote and by cueing from a calendar or schedule.
K.GM.B.5	Identify pennies, nickels, dimes and quarters.	The expectation of the student is to verbally name (identify) pennies, nickels, dimes and quarters; and identify these coins from pictures and manipulatives.
K.GM.C	**Analyze squares, circles, triangles, rectangles, hexagons, cubes, cones, cylinders and spheres.**	
K.GM.C.6	Identify shapes and describe objects in the environment using names of shapes, recognizing the name stays the same regardless of orientation or size.	The expectation of the student is to identify shapes (two-dimensional) and objects (three-dimensional) in their environment using the correct mathematical vocabulary and recognizing the name stays the same regardless of orientation or size.
K.GM.C.7	Describe the relative positions of objects in space.	The expectation of the student is to describe the relative positions of objects in space using terms such as *above, below, beside, in front of, behind* and *next to.*
K.GM.C.8	Identify and describe the attribute of shapes, and use the attributes to sort a collection of shapes.	The expectation of the student is to identify and describe, using the correct mathematical vocabulary, the attributes of two-dimensional and three-dimensional shapes, and using the attributes to sort a collection of shapes. Focus should be placed on measurable attributes.
K.GM.C.9	Draw or model simple two-dimensional shapes.	The expectation of the student is to draw or model (manipulatives, modeling clay, cut-outs, etc.) simple two-dimensional shapes (e.g. square, rectangle, rhombus, circle and triangle).

K.GM.C.10	Compose simple shapes to form larger shapes using manipulatives.	The expectation of the student is to compose simple shapes to form larger shapes using manipulatives. Draw or model (e.g. manipulatives, modeling clay and cut-outs) the composition of two-dimensional shapes (e.g. square, rectangle, rhombus, circle and triangle). *(e.g., "Can you join these two triangles with full sides touching to make a rectangle?")*
DATA AND STATISTICS: DS		
K.DS.A	**Classify objects and count the number of objects in each category.**	
K.DS.A.1	Classify objects into given categories; count the number of objects in each category.	The expectation for the student is to classify (sort) objects into given (or student generated) categories, defend the classification and count the number of objects in each category.
K.DS.A.2	Compare category counts using appropriate language.	The expectation of the student is to compare category counts based on a graphical representation using the correct mathematical vocabulary. (e.g., greater than, less than or equal to)